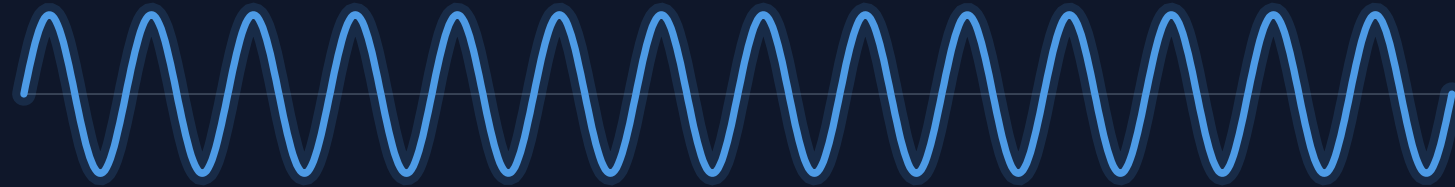


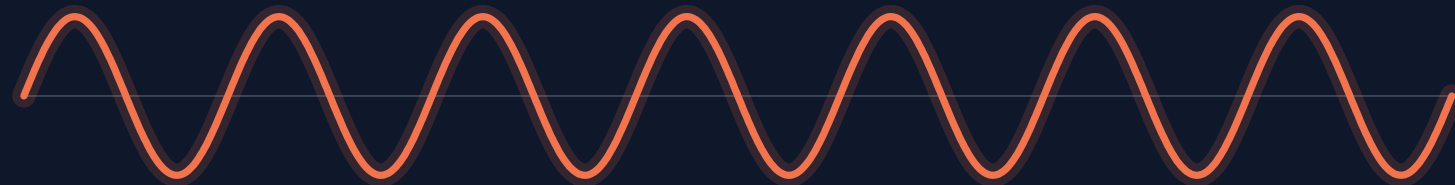
Cortical EEG slowing in resting-state Parkinson's disease

Final presentation: model, code, and findings

- **Control**



- **Parkinson's**



Hypotheses

RESEARCH QUESTION

how does resting EEG change in Parkinson's, and can one mechanism explain the slowing?

① theta power higher

in PD · $p < 0.001$

CONFIRMED

② alpha peak slows

9.25 → 8.00 Hz · $p < 0.001$

CONFIRMED

③ beta power lower

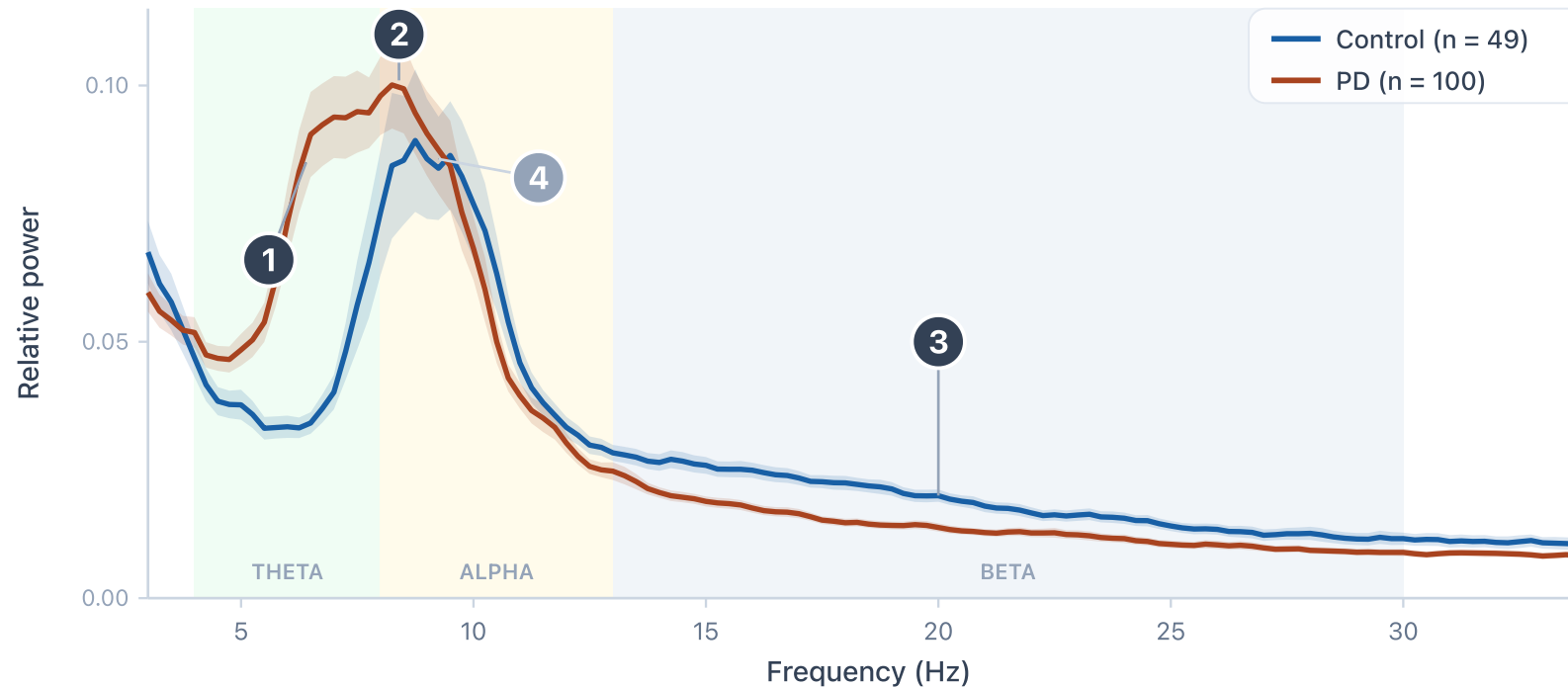
in PD · $p < 0.001$

CONFIRMED

④ alpha power

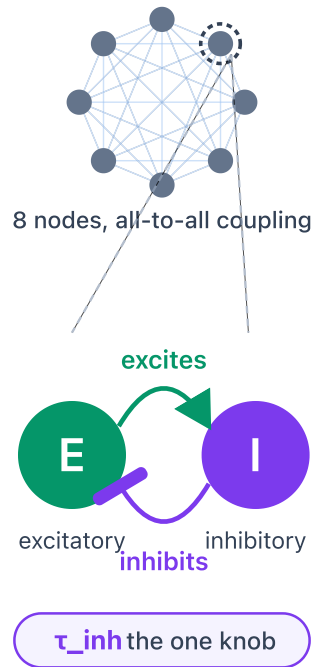
unchanged · $p = 0.9$

NULL



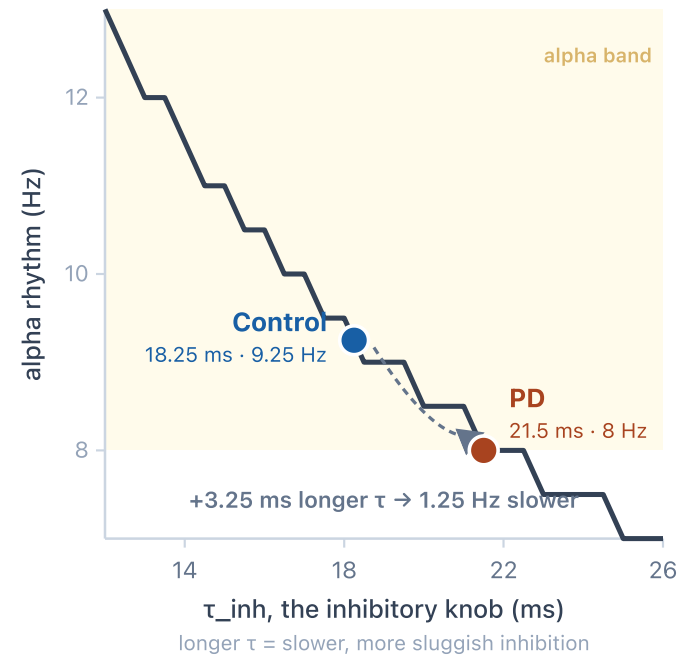
The model: a Wilson-Cowan network

THE MODEL: AN E-I LOOP

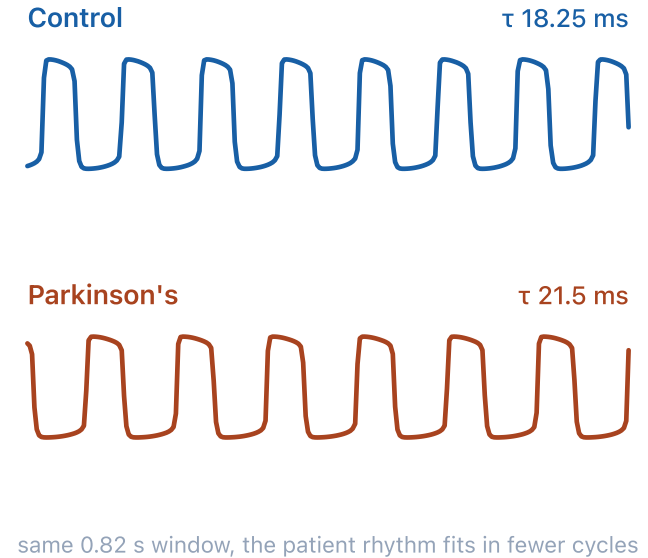


TURN τ_{inh} UP, THE RHYTHM SLOWS

each point = the model run at one knob setting



THE SIMULATED RHYTHM

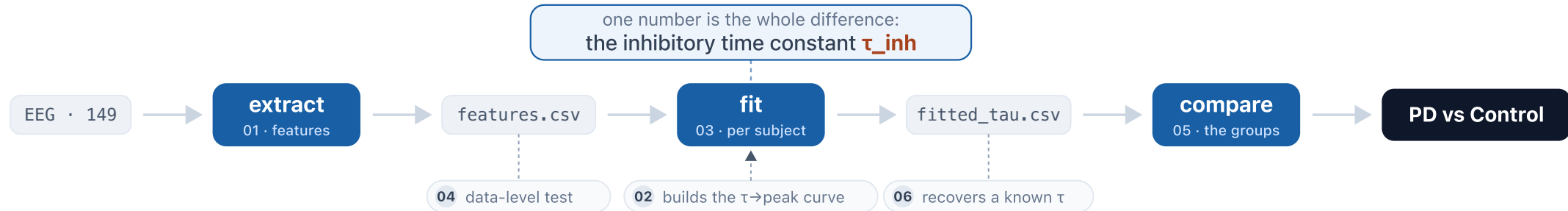


WHY WILSON-COWAN the rhythm comes out of the excitatory-inhibitory mechanism, so the slowing has a real cause we can name (slower inhibition), not just a frequency we dial down by hand. That cause is the one knob our hypothesis points to.

ONE KNOB ONLY every other setting is held fixed, so the whole change from healthy to patient is a single number, a longer inhibitory time constant τ_{inh} (about **+3.25 ms** in PD).

Code & repository

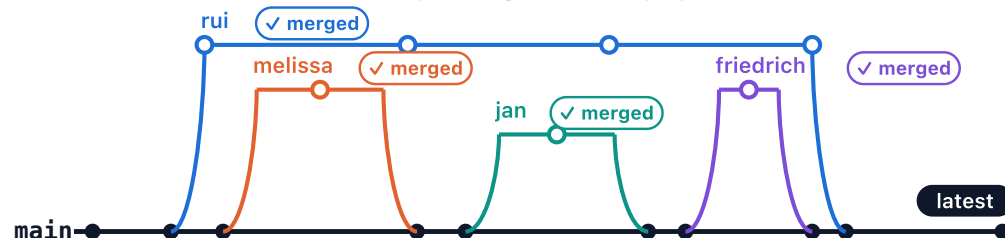
THE PIPELINE EEG to result, through six scripts



THE REPOSITORY



VERSION CONTROL 64 commits, all merged into main, reproducible



THE MODEL, IN CODE

one function, one free parameter

```

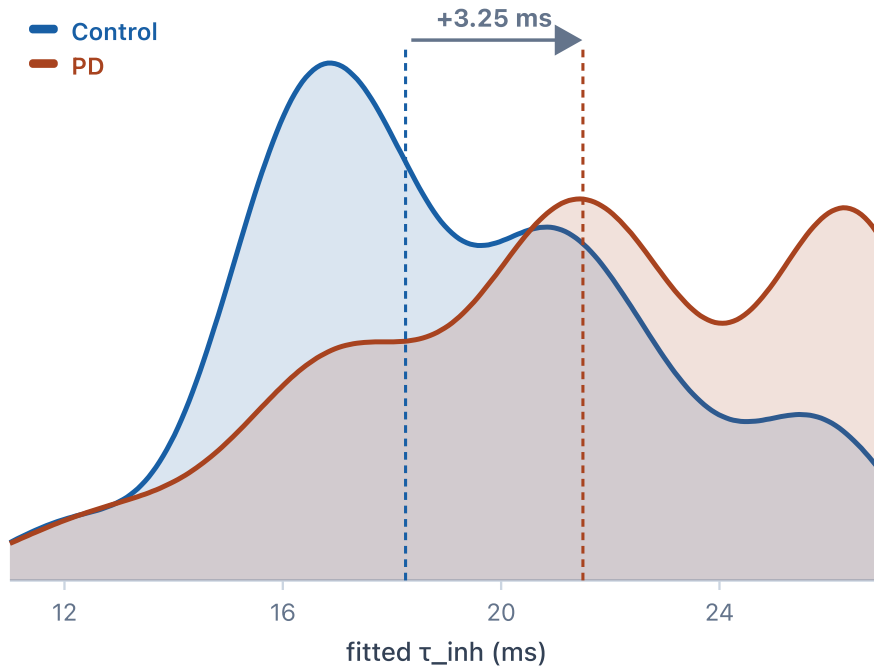
src/model.py

# the whole model is one function; one number is the difference
def build_network(n_nodes, tau_inh, *,
                  exc_ext=0.65, # external drive
                  sigma_ou=0, # deterministic, no noise
                  K_gl=0.6, # global coupling
                  duration_ms=8000): # simulation length
    """A Wilson-Cowan network. tau_inh is the one free
    parameter; everything else stays at these defaults."""
    Cmat, Dmat = all_to_all_coupling(n_nodes)
    model = WCModel(Cmat=Cmat, Dmat=Dmat)
    model.params["tau_inh"] = tau_inh # the one knob we turn
    model.params["exc_ext"] = exc_ext
    model.params["sigma_ou"] = sigma_ou
    model.params["K_gl"] = K_gl
    model.params["duration"] = duration_ms
    return model

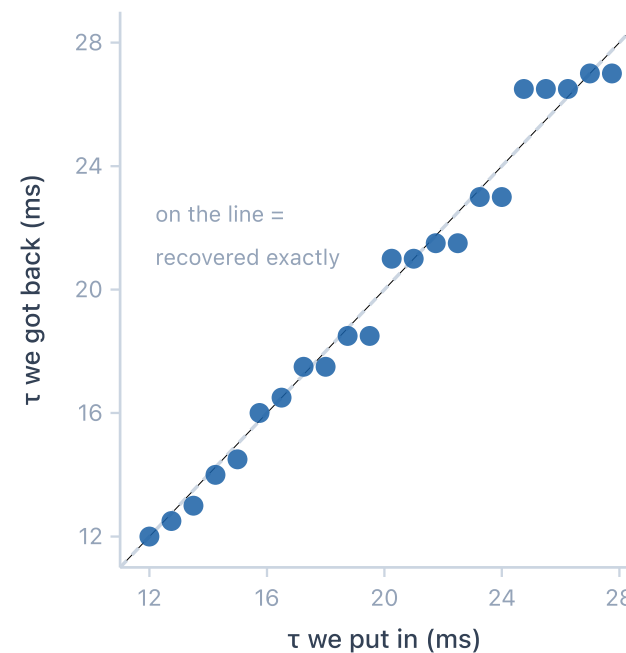
# healthy vs patient: same function, one number changed
control = build_network(8, tau_inh=18.25) # alpha 9.25 Hz
patient = build_network(8, tau_inh=21.50) # 8 Hz, +3.25 ms
  
```

Does the model separate patients from controls?

THE SEPARATION fitted τ_{inh} , Control vs PD



PARAMETER RECOVERY put a known τ in, fit it back



The fit is unbiased.

mean error -0.1 ms
typical error ± 0.5 ms

Fitted τ_{inh} re-expresses the measured slowing as a model parameter (one-sided Mann-Whitney $p \approx 0.0002$).

Outlook and open challenges

⚠ WHERE THE MODEL IS LIMITED



Alpha only

One parameter was tuned to the alpha peak, so theta and beta stay unexplained.



Describes, not proves

A different change could give the same slowing, so it re-describes the data, not the cause.



Coarse and uniform

Frequency is read in coarse steps, on a simplified, all-to-all network.

✅ WHERE IT GOES NEXT



Fit more than the peak

Match the whole spectrum or connectivity, and vary more, to reach theta and beta too.



Anchor it in biology

Tie the fitted inhibition to medication state (on vs off L-Dopa) and symptom severity.



Sharper and more realistic

Fit continuously, not from a lookup table, on a real whole-brain connectome.

What we took away

Rui

Carrying this end to end, from raw EEG to the fitted model, taught me that a result is only as trustworthy as the pipeline behind it; the reproducible scripts and the honest recovery check mattered as much as the model.

Melissa

My biggest learning was how much easier good documentation makes everything, whether it's the repository structure, script naming, clear responsibilities, or the code itself. It keeps the whole team aligned, helps everyone track progress, and makes even complex or unfamiliar methods feel more approachable.

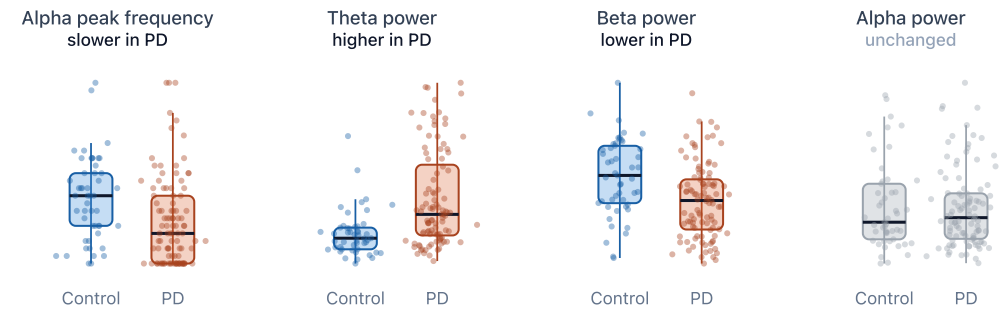
Jan

This course taught me that you do not have to be a master programmer to meaningfully contribute to, and clinically interpret, advanced computational research.

Friedrich (to come)

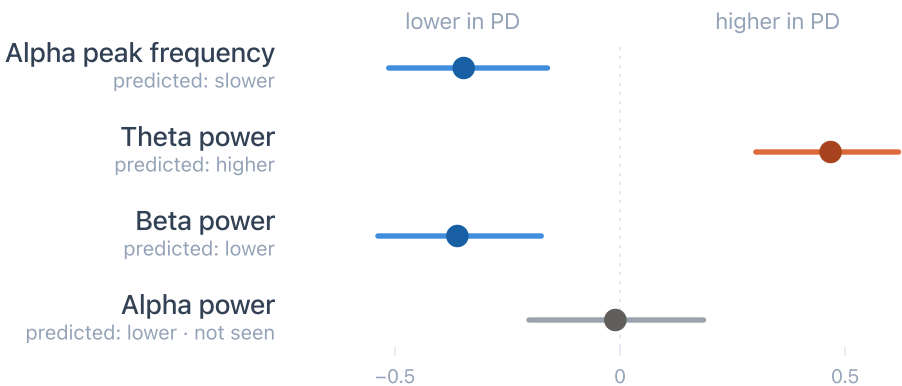
Appendix and references

PD vs Control: per-subject EEG features



How Parkinson's changes the resting EEG

effect size per feature, with 95% confidence interval



DATA Anjum MF, et al. Resting-state EEG measures cognitive impairment in Parkinson's disease. npj Parkinson's Disease (2024). doi:10.1038/s41531-023-00602-0

MODEL Wilson HR, Cowan JD. Excitatory and inhibitory interactions in localized populations of model neurons. Biophysical Journal 12, 1-24 (1972).

TOOL Cakan C, Jajcay N, Obermayer K. neurolib: a simulation framework for whole-brain neural mass modeling. Cognitive Computation (2023). doi:10.1007/s12559-021-09931-9